Solving Quadratic Equations

Exam code: 9MA0

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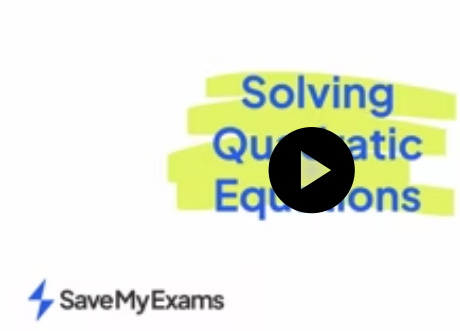
Updated on 19 June 2025



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Help me understand

Solving quadratic equations

Solving quadratic equations

- We can solve quadratic equations when they are written in the form $ax^2 + bx + c = 0$
- If given an unusual looking equation, try to rearrange it into this form first
- The three ways to solve a quadratic you must know are
  - Factorising
  - Completing the square
  - Quadratic formula

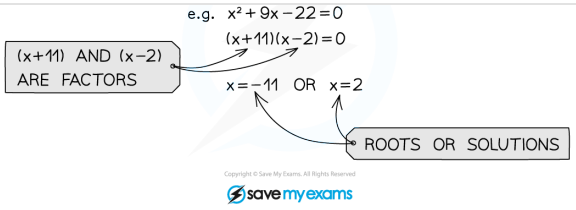
How do I solve by factorisation?

- Factorising is a great way to solve a quadratic quickly but won't work for all quadratics
- If the numbers are simple, try factorising first
- Once factorised, set each bracket to = 0 and solve

Stuck on something?

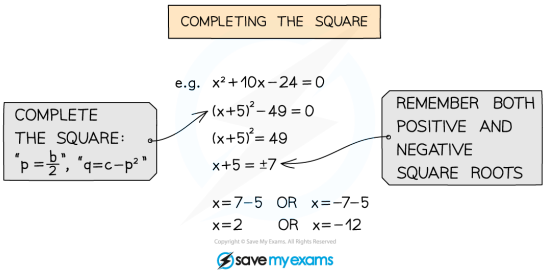
Ask questions and get help with exactly what you're revising.

Got it



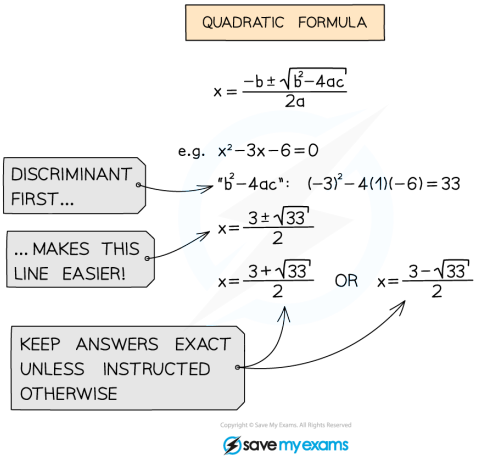
How do I solve by completing the square?

- Completing the square will work for any quadratic
- Make sure you know how to complete the square
- Remember this will help with questions involving turning points too

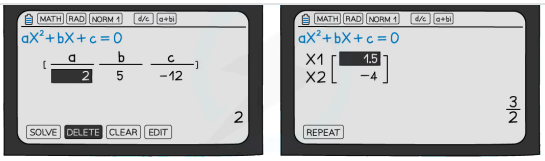


How do I solve using the quadratic formula?

- The quadratic formula might look complicated but it just uses the coefficients a, b and c from the quadratic equation
- The quadratic formula will work for any quadratic



How do I solve using a calculator?



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- So the solution to the quadratic equation $2x^2 + 5x - 12 = 0$ are **x = 1.5** and **x = -4**

Examiner Tips and Tricks

- A calculator can be super-efficient but be aware some marks are for method
- There will never be many marks for solving a quadratic at AS/A level
- Use your judgement:
  - is it a "show that" or "prove" question?
  - how many marks?
  - how long is the question?
- Remember the quadratic formula with a song... there are loads of fun ones on YouTube

Worked Example

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